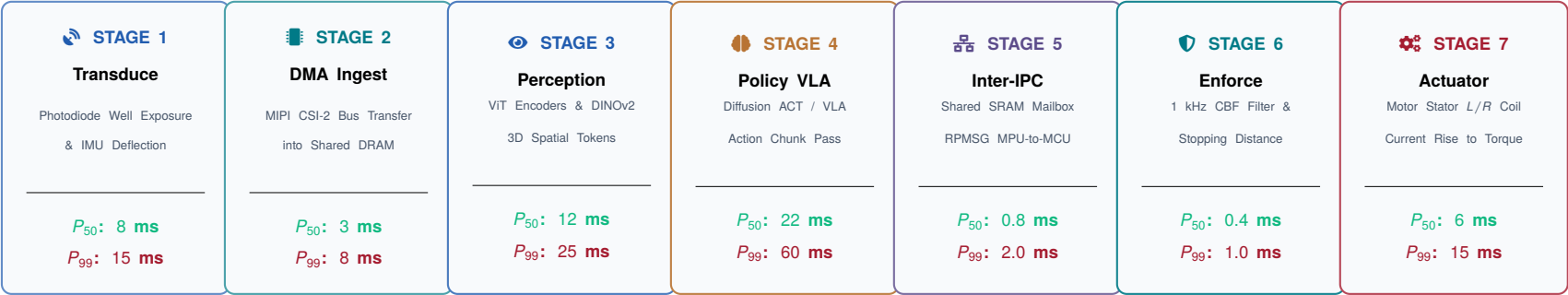




THE 7-STAGE SENSE-TO-ACTUATION LATENCY WATERFALL

Where Does the Time Go? Deconstructing the Physical, Memory, Neural, and Silicon Bottlenecks



✓ **Nominal Path** ($P_{50} = 52.2$ ms):

⚠ **Tail Path** ($P_{99} = 126.0$ ms):

Safe Closed-Loop Margin (52.2 ms $\ll \tau_{\text{world}}$)

CRITICAL TIMING VIOLATION (126.0 ms $> \tau_{\text{world}}$)

World Deadline $\tau_{\text{world}} = 100$ ms